

Assignment8

2025-10-29

```
install.packages("readxl")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library(readxl)
install.packages("ggplot2")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library(ggplot2)
install.packages("dplyr")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

install.packages("GGally")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library("GGally")
install.packages("tidyverse")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.1      v stringr 1.5.2
## v lubridate 1.9.4    v tibble 3.3.0
## v purrr 1.2.0       v tidyr 1.3.1
## v readr 2.1.6

## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
```

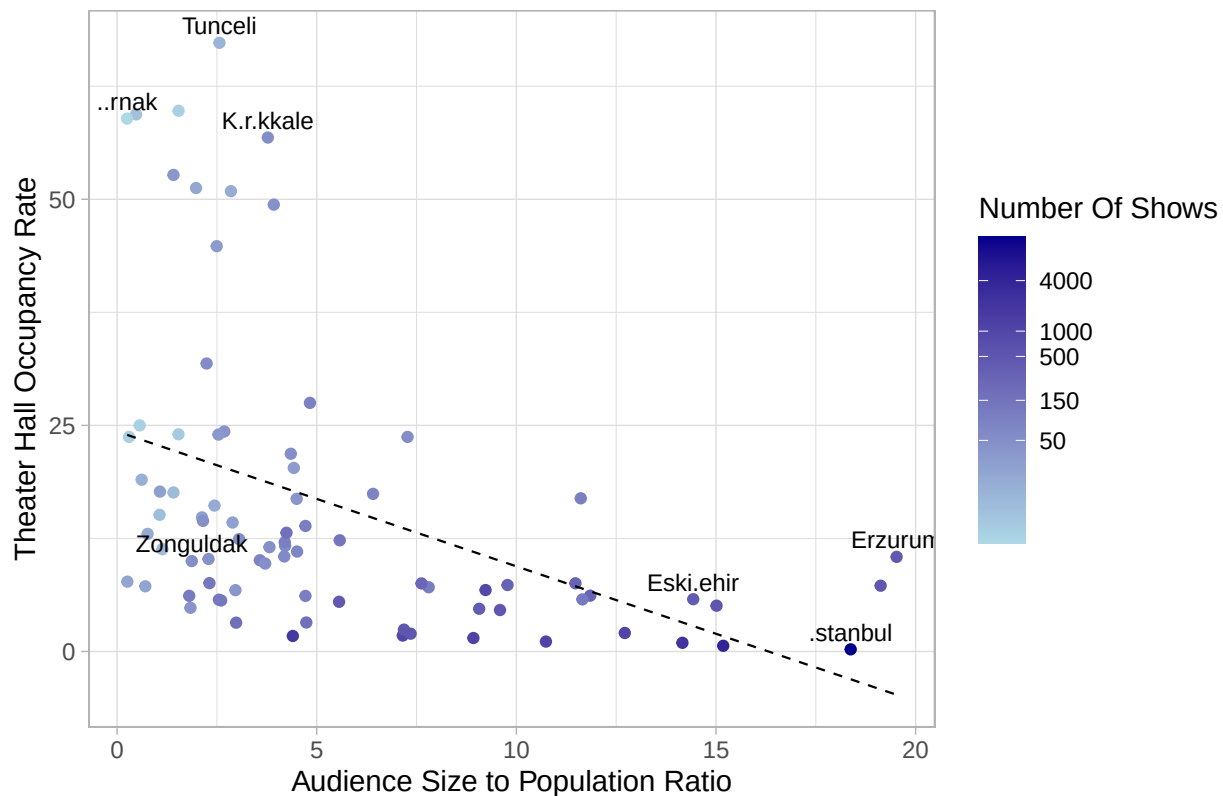
```
## x dplyr::lag()      masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
data<-read_excel("theatre_data.xlsx")
```

```
etiket<- data %>%
  filter(citys %in% c("İstanbul","Zonguldak","Şırnak","Tunceli","Eskişehir","Kırıkkale","Erzurum"))

ggplot(data, aes(x = ap_rate, y = s_rate, colour = perf)) +
  geom_point() +
  geom_text(data=etiket,aes(label=citys),
            vjust=-0.5,
            hjust=0.5,
            size=3,
            color="black")+
  geom_smooth(method=lm , color="black", se=FALSE, linetype="dashed",linewidth=0.4)+
  labs(title = "Correlation Between Theater Attendance-Hall Occupancy Rate By Province",
       x="Audience Size to Population Ratio",
       y="Theater Hall Occupancy Rate",
       colour="Number Of Shows")+
  scale_color_gradient(low="lightblue",high="darkblue",trans="log10",breaks=c(50,150,500,1000,4000))+
  guides(color=guide_colorbar(barheight=8))+
  theme_light()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Correlation Between Theater Attendance-Hall Occupancy Rate By Province



```
k <- max(data$y_perf, na.rm = TRUE) / max(data$y_ap_rate, na.rm = TRUE)
ggplot(data, aes(x = year)) +
```

```

geom_line(aes(y = y_perf, color = "Total Number Of Shows"), linewidth = 1,alpha=0.5) +
geom_line(aes(y = y_ap_rate * k, color = "Audience To Population Ratio"), linewidth = 1,alpha=0.5) +
scale_y_continuous(
  name = "Total Number Of Shows",
  sec.axis = sec_axis(~./k, name = "Audience To Population Ratio")) +
scale_color_manual(name = "Variables", values = c("Total Number Of Shows" = "darkcyan", "Audience To
labs(title = "Time Series Comparison",
  x="Year") +
theme_minimal() +
theme(
  panel.border = element_blank(),
  panel.grid.major.x = element_blank(),
  panel.grid.minor.x = element_blank(),
  panel.grid.major.y = element_line(color = "azure4", linetype = "solid",size=0.5),
  panel.grid.minor.y = element_blank(),
  axis.line.x = element_line(color = "black"),
  axis.line.y = element_blank(),
  axis.title.y.left = element_text(color = "darkcyan"),
  axis.title.y.right = element_text(color = "deeppink"),
  axis.text.x = element_text(angle = 45, hjust = 1)
)

```

```

## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

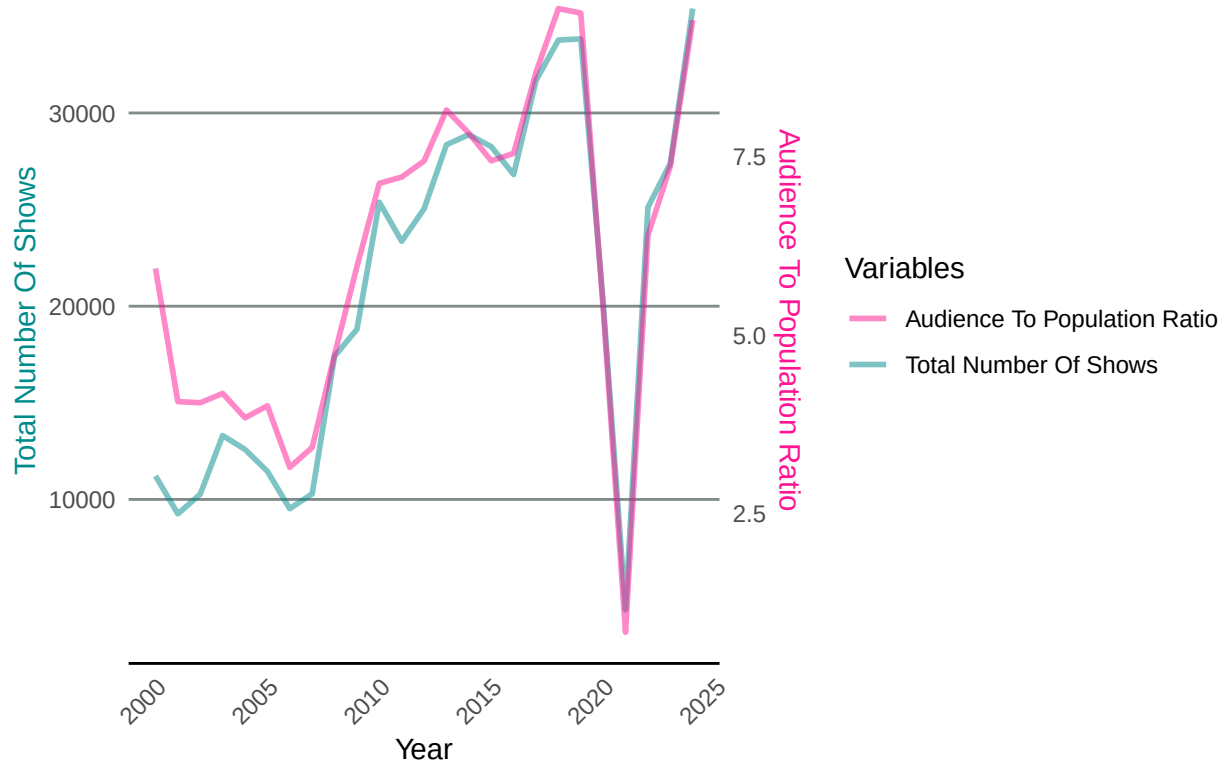
```

```

## Warning: Removed 56 rows containing missing values or values outside the scale range
## (`geom_line()`).
## Removed 56 rows containing missing values or values outside the scale range
## (`geom_line()`).

```

Time Series Comparison



```
install.packages("devtools")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```
library(devtools)
```

```
## Loading required package: usethis
```

```
devtools::install_github("htastan/TRmaps")
```

```
## Skipping install of 'TRmaps' from a github remote, the SHA1 (c136d9fa) has not changed since last in
## Use `force = TRUE` to force installation
```

```
library(TRmaps)
```

```
install.packages("sf")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```
library(sf)
```

```
## Linking to GEOS 3.8.0, GDAL 3.0.4, PROJ 6.3.1; sf_use_s2() is TRUE
```

```
data("tr_nuts3")
```

```
data2 <- left_join(tr_nuts3, data, by = c("name_tr" = "citys"))
```

```
ggplot(data2) +
  geom_sf(aes(fill = ap_rate)) +
  geom_sf_text(aes(label=name_tr),size=1.5)+
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```